Econometrics Empirical Project Report

Poverty and Obesity

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Abstract

It is common knowledge that obesity rates have risen significantly in the past few decades, causing a myriad of health problems in the affected regions. Our aim is to test the effect of various factors on obesity, focussing primarily on poverty. At first we used a single regression model to compare country GDP and average population BMI in 162 different countries with the assumption that the two will be positively correlated. Then we focussed on developed countries like the USA to find the correlation between obesity, poverty, and factors such as education levels using a single and multiple regression model. We found significance between per capita income and prevalence of obesity in the United States in our single regression model at all levels, but were unable to find meaningful data correlation in our multiple regression; however, we did find a very significant negative impact of high school graduation rates on obesity rates and a positive correlation between the percentage of non-Hispanic white population and obesity rates by state.

1. Introduction

BMI (Body Mass Index) is a crude measure of obesity measured by dividing a person's weight by the square of their height. A BMI of 30 or more is considered obese, and a BMI of 25 or greater is considered overweight. It is known that as average population BMI rises so too does the prevalence of diseases and disorders like diabetes, cardiovascular diseases, and various types of cancer. In 2010 obesity and overweight were estimated to be the leading cause of at least 3.4 million deaths worldwide. The rise in obesity-related mortality rates has also contributed to a rise in overall medical expenditures, placing a growing financial burden on health services and the economy overall.

It has been established that increasingly easy access to calorically dense and nutrient poor food has been a large factor in the rising rates of obesity in the developed world and countries with a higher GDP. Cheap, high calorie foods have become a less expensive and sometimes easier alternative to cooking healthy meals thanks to the rise and proliferation of fast food throughout the developed world. Because of this we expect that due to the relatively high cost of healthy living, there will be a correlation between poverty and obesity rates in the developed world.

In countries where there is not such easy access to calorically dense foods or even food in general we assume there will be an opposite trend in obesity rates. We hypothesize that in these poorer countries there will be a generally lower BMI as compared with countries with larger GDP. This rationale provides the basis for our single regression models.

2. Data

This paper focuses on worldwide and national trends in Obesity. Worldwide, the study focuses on the impact of a country's GDP on their average BMI. Nationally, the study focuses on the impact of each state's per capita income on their average BMI.

2.1 Single regression

We are checking whether or not GDP has an effect on average country BMI in order to find whether the data supports our hypothesis that these two are positively correlated. The model is as follows:

$$BMI = \beta_0 + \beta_1 GDP + u$$

GDP per Capita vs BMI

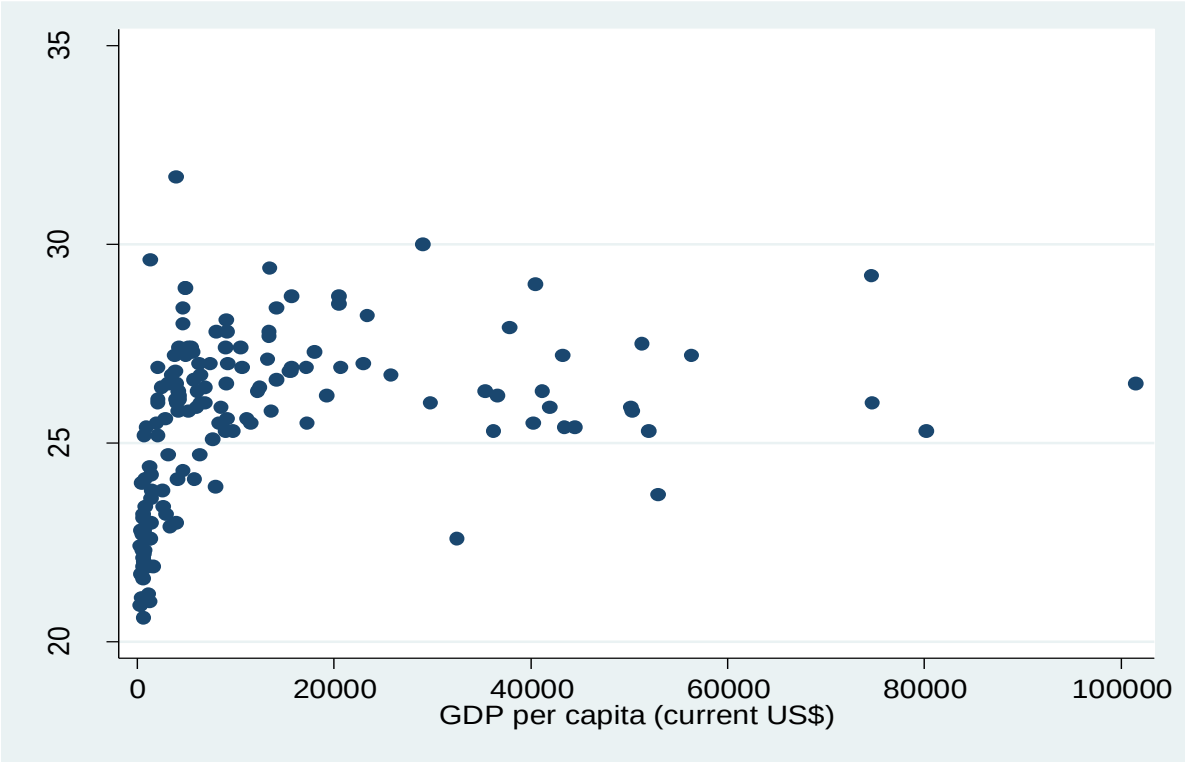
Source	SS	df	MS
Model	67.2591497	1	67.2591497
Residual	641.767243	148	4.33626515
Total	709.026392	149	4.75856639

Number of obs =	150
F(1, 148) =	15.51
Prob > F =	0.0001
R-squared =	0.0949
Adj R-squared =	0.0887
Root MSE =	2.0824

meanbmi	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
gdppercapitacurrentus	.0000372	9.46e-06	3.94	0.000	.0000186	.0000559
_cons	25.04447	.2103153	119.08	0.000	24.62886	25.46007

The dependent variable in this case is average country BMI with GDP as our independent variable. We chose this independent variable to test the correlation between larger GDP and higher average BMI. We used data of average BMI by country from the World Health Organization and data for country GDP from the World Bank.

Based on the following table and the scatter plot below we can conclude that GDP per capita is significant at all levels and we found a generally positive correlation between average country BMI and GDP.



2.2 Second Single Regression

Our second regression follows the previous in regards to the impact of income/poverty on obesity. Through this model we are trying to find out whether or not state per capita income has an effect on the prevalence of obesity. We utilized this regression in order to find whether or not the data supports our hypothesis that these two factors are negatively correlated.

The dependent variable in this case is the prevalence of obesity by state, with state per capita income as our independent variable. We chose this independent variable to test the existence of a negative correlation between larger income and a greater prevalence of obesity.

We used data of obesity prevalence by state from the CDC and data for state per capita income from the United States Census Bureau. We were able to find data points for all 50 states in the United States, plus the District of Columbia.

The model is as follows:

$$Obesity\ Rate = \beta 0 + \beta 1 * Per\ Capita\ Income + \mu$$

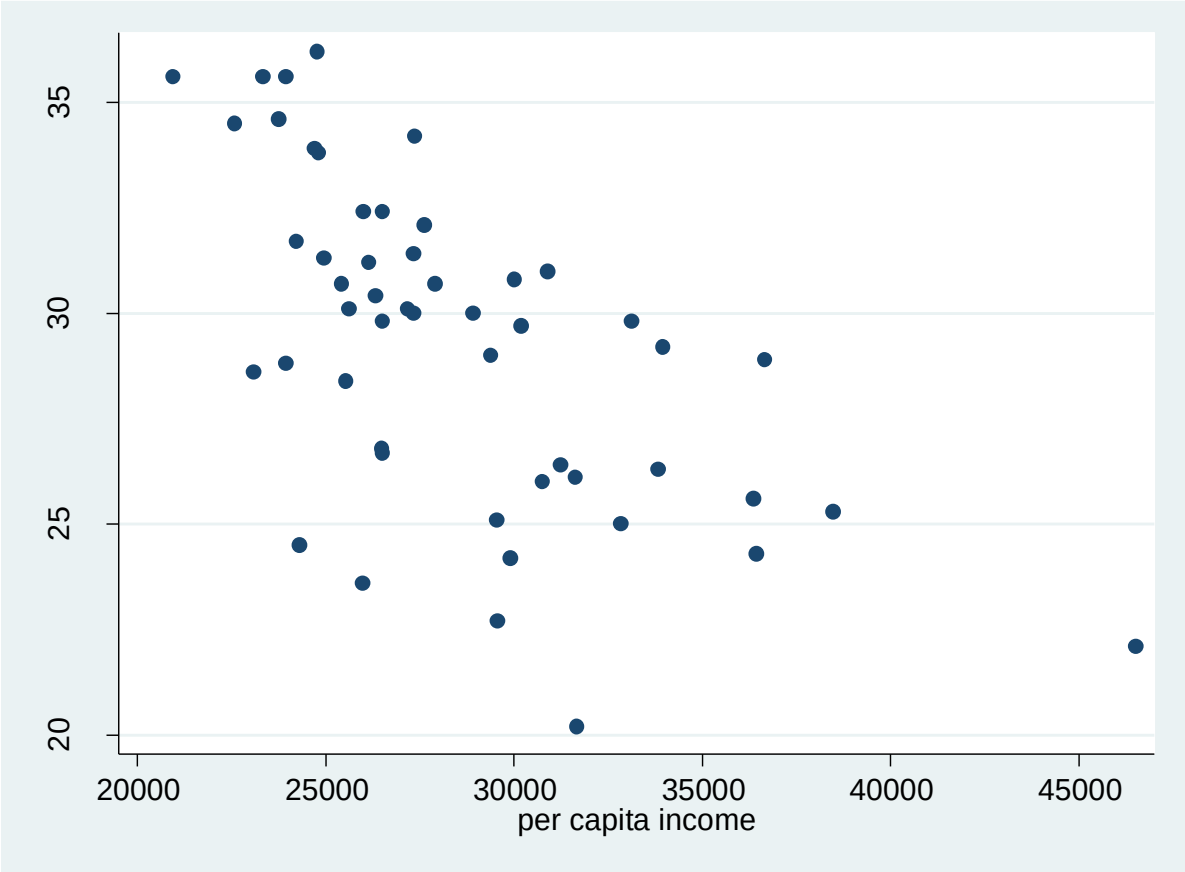
The dependent variable in this case is the prevalence of obesity by state. The state per capita income is our independent variable.

Source	SS	df	MS
Model	286.112944	1	286.112944
Residual	463.06111	49	9.45022673
Total	749.174054	50	14.9834811

Number of obs =	51
F(1, 49) =	30.28
Prob > F =	0.0000
R-squared =	0.3819
Adj R-squared =	0.3693
Root MSE =	3.0741

obesitypercent	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
percapitaincome	-.0005017	.0000912	-5.50	0.000	-.000685	-.0003185
_cons	43.57718	2.633373	16.55	0.000	38.28522	48.86914

Scatter Plot



Based on the regression table and the scatter plot we found that Capital Income is significant at all levels, which shows that there is an impact of per capita income on obesity prevalence. But, the low R square value suggests that

capital income doesn't explain much of the variation in obesity prevalence. Therefore, we decided to do a multiple regression model considering the impact of other factors on the prevalence of obesity by state.

2.3 *Multiple Regression*

For our multiple regression model we chose to focus on a single country: the United States. This model looks at the impact of per capita income, age, high school graduation rate, health-care spending, and percentage of non-Hispanic whites on the prevalence of obesity by state. Data for average income level, age, percent of population that had graduated high school, and percentage of the population that are non-Hispanic white came from the United States Census Bureau (census 2010).

Source	SS	df	MS
Model	298.477216	5	59.6954433
Residual	291.510235	45	6.47800521
Total	589.987451	50	11.799749

Number of obs =	51
F(5, 45) =	9.22
Prob > F =	0.0000
R-squared =	0.5059
Adj R-squared =	0.4510
Root MSE =	2.5452

PrevObesity	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
CapitaIncome	-.0000974	.0000811	-1.20	0.236	-.0002606	.0000659
Age	.0096907	.2090841	0.05	0.963	-.4114263	.4308077
HighSchool	-67.24224	13.01327	-5.17	0.000	-93.45231	-41.03218
HealthSpend	-8.14e-06	.0004342	-0.02	0.985	-.0008826	.0008663
PercWhite	.1231913	.0271512	4.54	0.000	.0685059	.1778767
_cons	80.79419	11.59884	6.97	0.000	57.43293	104.1555

Our first independent variable is average income level, based on our hypothesis that it would be negatively correlated with obesity prevalence. Age was chosen as an explanatory variable based on the assumptions that, generally, as people age their lifestyles become more sedentary. High school graduation rate was a relatively intuitive choice based on the idea that a better educated population would make smarter nutrition decisions. Healthcare spending was an obvious choice due to the general increase in the cost of obesity related diseases and ailments, and percentage non-Hispanic white population was used to determine cultural/racial effects on obesity.

	PrevOb~y	Capita~e	Age	HighSc~l	Health~d	PercWh~e
PrevObesity	1.0000					
CapitaIncome	-0.2743	1.0000				
Age	0.0561	0.0323	1.0000			
HighSchool	-0.4519	0.2642	0.2495	1.0000		
HealthSpend	-0.2097	0.1120	0.5040	0.3489	1.0000	
PercWhite	0.2936	0.0546	0.3816	0.4225	0.0612	1.0000

As can be seen in the Table above, the independent and dependent variables are not perfectly collinear, and as such, do not violate the Gauss-Markov Theorem.

The results for the multiple regression using STATA are as follows:

*PrevObesity = 80.79 - .000097(CapitaIncome) + .0097*Age - .6724(Highschool) - 8.14e-6(HealthSpend) + .12(PercWhite)*

The new independent variables included in the model are average age, percentage of high school graduates, health expenditure, and percentage of the non-Hispanic white population. With p-values of .236, .963, and .985 respectively, income per capita, age and health care expenditure were shown not to be statistically significant. All three of these aforementioned p-values are far greater than .10, .05, and .01, and as such they are statistically insignificant at the 10%, 5%, and 1% levels., However, the percentage of high school graduates and the percentage of non-Hispanic white population were shown to be statistically significant at the 1%, 5%, and 10% levels as both independent variables had a p-value of 0.

Since the income per capita, the average age of the population and the health care expenditure were shown to be statistically insignificant at the 1%, 5%, and 10% levels, we then did a regression without these three variables. The third regression model listed reflects the restricted regression model, which removed the mentioned variables. After using the f-statistic test, the three removed variables were shown to also be jointly insignificant: the f-value obtained was .436, which does not exceed the f-value needed to be statistically significant.

F = [($SSRr$) - ($SSRur$)]/q÷ [($SSRur$)/(n-k-1)] = [(300)-(291.51)]/3÷[((291.51)/45]
F = .436

Therefore, the model that best predicts obesity rates would be the restricted multivariable regression model. The independent variables that were removed were found to also be jointly insignificant, thus we can use a model that does not include these variables at all. It is important to test for joint significance because if the variables were found to be significant together, then they would still be important for the model.

3. Conclusions

The lack of significance of *CapitaIncome* on *PrevObesity* in our multiple regression model led us to further investigate the other independent variables in our multiple regression. In doing so we found that the percentage of high school graduates in a state as well as the non-hispanic white population were significant at the 10%, 5%, and 1% level. *Highschool* had a negative impact on the prevalence of obesity while *percwhite* had a positive impact. We believe that the lack of significance of *CapIncome* on *PrevObesity* has to do with the similar level of access to food throughout all income levels in the United States. Generally low cost foods in the United States are more calorically dense while being less nutrient rich (McDonalds and other fast food) leading the poorer population to rely on these food sources, while the richer population is free to overeat more varied food types.

References:

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